

JARED THRONE

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EDUCATION

Michigan State University, College of Engineering

May 2025

Bachelor of Science, Mechanical Engineering | GPA: 3.5 / 4.0

PROFESSIONAL EXPERIENCE

Fluids Systems Engineer

February 2026 – Present

Quest Defense | Blue Ash, OH

- Own design of fuel and hydraulic components (pumps, actuators, mechanical controls) for Fortune 500 customers; capture and incorporate system requirements into component designs.
- Manage suppliers across cost, performance, and manufacturability; design component interfaces and perform tolerance stack-up analysis.
- Present in technical and program reviews, interfacing directly with internal and external engineering groups to resolve complex design problems.

Research Intern

May – August 2024

RWTH Aachen University | Aachen, Germany

- Executed time-resolved PIV experiments across four vessel geometries using blood analogous fluid; analyzed data with MATLAB and CATIA.
- Simulated internal flow using ANSYS Fluent to validate experimental thermofluid findings across vessel geometry variations.

Systems Engineering Intern, Thermal and Process Control

May – October 2023

Fraunhofer USA CMW | East Lansing, MI

- Designed water conditioning systems achieving 25 GPM flow rate for diamond growth thermal regulation under strict design requirements.
- Interpreted GD&T; drawings for complex assemblies and developed safety systems improving workplace safety for 20+ employees.

Process and Equipment Technician

November 2023 – May 2024

Great Lakes Crystal Technologies | East Lansing, MI

- CNC machined molybdenum components for high-spec manufacturing environments, maintaining stringent dimensional and quality standards.
- Coordinated cross-functional troubleshooting to resolve equipment and process issues, minimizing production downtime.

PROJECTS

NASA Psyche Mission Rover

Dec 2024

Designed rover mechanics in SolidWorks for asteroid 16-Psyche terrain; performed FEA, thermal, and structural simulation for component validation and rover dynamics.

Mini Segway, ME 456

Apr 2025

Tuned a PID controller in MATLAB Simulink to navigate a self-balancing segway through a competitive obstacle course design challenge.

MAC Launcher, ME 470

Apr 2024

Engineered a cornhole launcher using iterative design, FEA, and manufacturability constraints to optimize accuracy and ease of fabrication.

Vortex Shedding Visualization, ME 332

Apr 2024

Ran controlled flow-structure experiments with hydrogen bubble visualization; analyzed Reynolds and Strouhal number relationships.

TECHNICAL SKILLS

CAD: Siemens NX | SolidWorks | CATIA | AutoCAD **Simulation:** ANSYS Fluent | FEA | Thermal Analysis | MATLAB | Simulink

Methods: GD&T; | Tolerance Stack-Up Analysis | Design for Manufacturability | Experimental Design **Programming:** MATLAB | Python

Manufacturing: CNC Machining | Milling | Additive Manufacturing | 3D Printing | Mechanical Assembly